LESSON PLAN

SFNA (MOD AIRMANSHIP) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Type of grease and lubricants and their uses
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Type of grease and lubricants and their uses
5. Specific Objectives	24.1.3.1 Trainee will be able to identify various types of greases, fluids and compounds.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of grease types using sample kits; Safety: Wear gloves and eye protection; Supervision: Instructor monitors sample handling
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a lubricant in an aircraft engine?
(b) Name one common type of grease used in aviation maintenance.
(c) Why is it critical to select the correct lubricant for specific aircraft parts?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Classification of Aviation Greases • Lithium Complex Grease Properties • Synthetic Grease Applications • Selection Criteria for Greases	L & Ds	PPT, IB	15 min
(b)	• Hydraulic Fluid Types and Viscosity • Anti-Corrosion Compound Usage • Deicing and Anti-Freezing Compounds	L & Ds	PPT, PDF	20 min
(c)	• Coolant Liquids and Water Mixtures • Lubricant Storage and Handling • Contamination Prevention Measures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between lithium complex and synthetic greases based on operating temperature.
(b) Describe the specific use case for anti-corrosion compounds on aircraft metal surfaces.
(c) How does improper selection of hydraulic fluid affect aircraft system performance?

12. Home Assignment and Follow Up	Prepare a brief report comparing the properties of standard hydraulic fluids with those used in water injection mixtures, referencing the next lesson plan (LP 05).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Various types of hydraulic fluids, anti-corrosion compounds, deicing and anti-freezing compounds, coolant liquids, water injection mixtures, cooling oils, transformer oils, flushing oil, corrosion inhibitors etc.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min